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CLASS ACTIVITY – 01
ITA 0609 MACHINE LEARNING

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1. Student Failure Prediction System

Problem Statement

A university wants to identify students who are likely to fail a course before the semester ends using academic performance and attendance data.

Machine Learning Solution

Machine learning can help universities identify students who are at risk of failing by analysing their academic records and attendance patterns. Instead of waiting until the final examination, the system predicts students who may struggle in the course, allowing faculty members to provide timely support through counselling, remedial classes, or additional mentoring.

- **Objective:** Predict whether a student is likely to pass or fail before the semester ends.
- **Data Collection:** Attendance percentage, internal assessment marks, assignment scores, quiz marks, previous semester GPA, class participation, and assignment submission history.
- **Machine Learning Model:** A **Random Forest Classifier** is suitable because it provides high accuracy and handles multiple features effectively.
- **Training Process:** Train the model using historical student records where the final outcome (Pass/Fail) is already known.
- **Evaluation Metrics:** Accuracy, Precision, Recall, and F1-Score.

Expected Output: The system classifies students as **Low Risk**, **Medium Risk**, or **High Risk** of failing.

Student ID	Attendance	Internal Marks	Prediction	Risk Level
S101	92%	85	Pass	Low Risk
S102	68%	48	Fail	High Risk
S103	78%	60	Pass	Medium Risk

2. Traffic Congestion Prediction System

Problem Statement

A smart city authority wants to predict traffic congestion at major intersections one hour in advance using traffic cameras, GPS devices, and weather reports.

Machine Learning Solution

Traffic congestion can be predicted by analysing real-time traffic data along with weather conditions and historical traffic patterns. A machine learning model continuously learns from previous traffic behaviour and forecasts congestion levels one hour in advance. These predictions help city authorities optimize traffic signals and guide commuters through alternative routes.

- **Objective:** Predict traffic congestion one hour before it occurs.
- **Data Collection:** Vehicle count from traffic cameras, GPS speed and location, weather reports, time of day, day of the week, holidays, and accident information.
- **Data Preprocessing:** Remove missing values, aggregate data into hourly intervals, normalize numerical values, and generate time-based features.
- **Machine Learning Model:** An **LSTM (Long Short-Term Memory)** network is suitable because it performs well with time-series data. XGBoost is another effective alternative.
- **Training Process:** Train the model using historical traffic and weather data collected over several months.
- **Evaluation Metrics:** Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction accuracy.

Expected Output: The model predicts whether traffic conditions will be **Low**, **Moderate**, or **Heavy** one hour in advance.

Intersection	Current Traffic	Predicted Traffic (1 Hour Later)	Congestion Level
Junction A	Moderate	Heavy	High
Junction B	Low	Moderate	Medium
Junction C	Heavy	Heavy	High

3. Defective Product Detection System

Problem Statement

A manufacturing company wants to automatically identify defective products from images captured on the production line.

Machine Learning Solution

Manual inspection of products is often time-consuming and may lead to human errors. A machine learning-based image classification system can automatically analyse product images captured during manufacturing and detect defects with high accuracy. This improves quality control while reducing inspection time.

- **Objective:** Automatically identify defective products from production-line images.
- **Data Collection:** Images of defective and non-defective products collected from the manufacturing process.
- **Data Preprocessing:** Resize images, normalize pixel values, and apply image augmentation techniques such as rotation, flipping, and brightness adjustment.
- **Machine Learning Model:** A **Convolutional Neural Network (CNN)** such as **ResNet50** or **Mobile Net** is suitable for image classification tasks.
- **Training Process:** Train the CNN using labelled product images so it learns the visual characteristics of defective and non-defective products.
- **Evaluation Metrics:** Accuracy, Precision, Recall, and Confusion Matrix.

Expected Output: The system classifies every product as **Defective** or **Non-Defective**.

Product ID	Prediction	Confidence
P001	Non-Defective	98%
P002	Defective	96%
P003	Non-Defective	97%